

1. **Define shoring and describe its type with neat and clean sketches. [3+7]**

Answer:

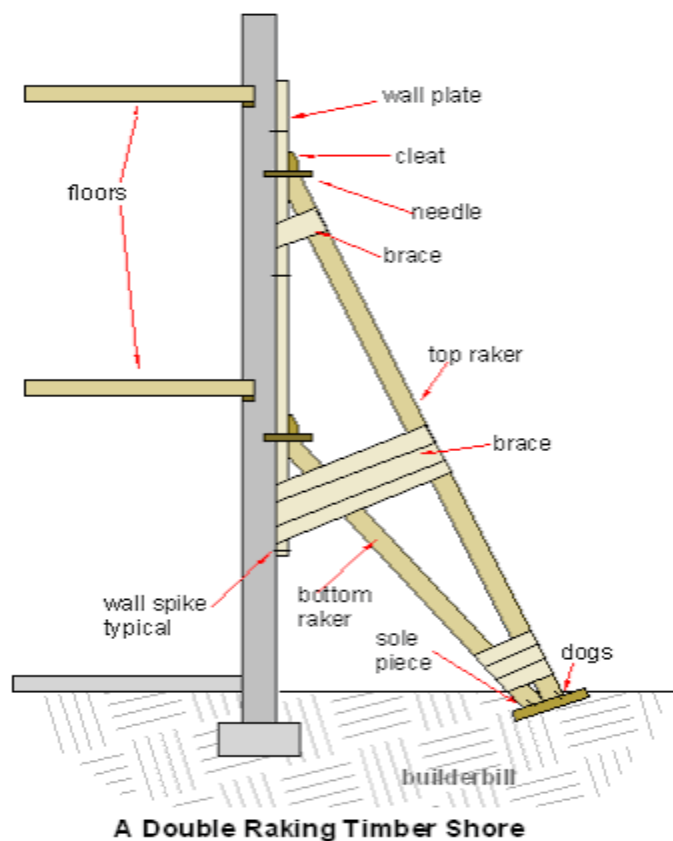
Shoring is a temporary structural arrangement provided to support unsafe or weak walls, buildings, trenches, or excavations to prevent their collapse during construction, repair, alteration, or demolition work.

Types of shoring:

The main types of shoring are:

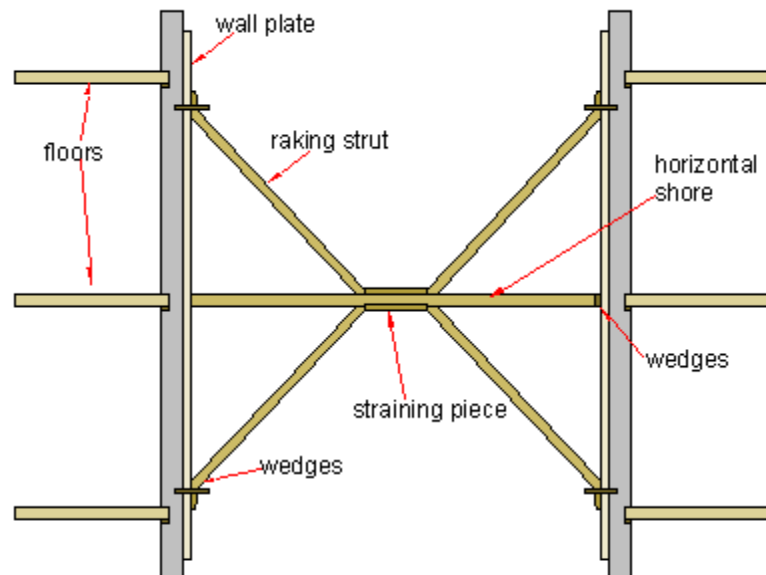
i. **Raking Shore**

- Consists of inclined members called rakers placed against the wall.
- The rakers transfer the wall load safely to the ground.
- Used when an external wall becomes unsafe or tends to bulge outward.



ii. **Flying Shore**

- Consists of horizontal or slightly inclined members connecting two parallel walls.
- It provides temporary support without obstructing the space between the walls.
- Used when an intermediate building is removed or when one wall needs support.

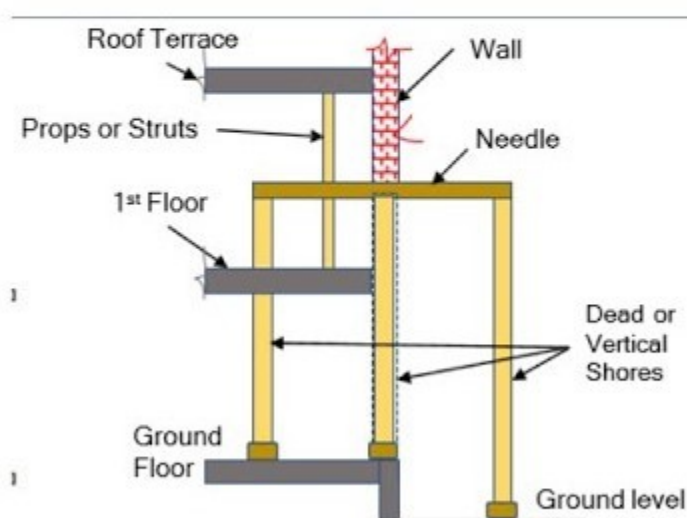


**Part Elevation Through Two Buildings
With A Single Flying Timber Shore**

iii. **Dead Shore**

- Consists of vertical posts supporting the walls, floors, roofs, or beams directly.
- Used when the lower portion of a wall is removed or openings are made in an existing wall.
- It transfers the load vertically to the ground.'

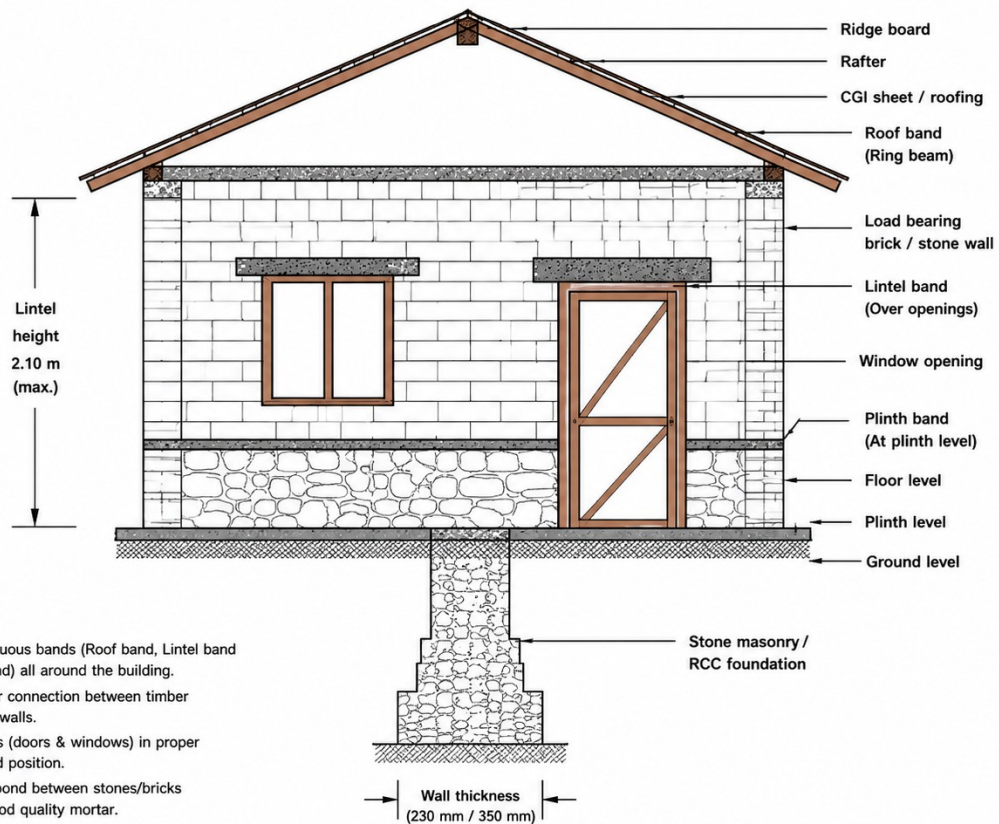
DEAD SHORES



2. **Draw a typical wall section of an earthquake resistant single storey building with load bearing walls.**

Answer:

**TYPICAL WALL SECTION OF AN EARTHQUAKE RESISTANT
SINGLE STOREY BUILDING WITH LOAD BEARING WALLS**



3. **Write briefly the bonds in brickwork and describe English bond with sketches of alternate layers and isometric view.**

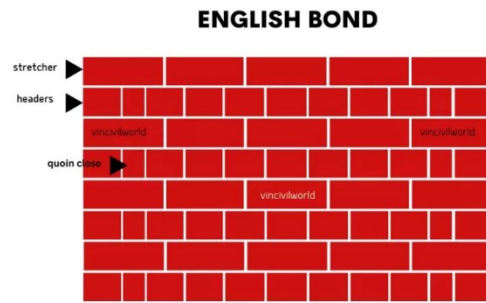
Answer:

Bond is the systematic arrangement of bricks in successive courses so that the bricks interlock properly, act as a single structural unit, and provide adequate strength and stability.

Common types of brick bonds:

- i. Stretcher bond
- ii. Header bond
- iii. English bond
- iv. Flemish bond
- v. Garden-wall bond
- vi. Dutch bond
- vii. Facing bond
- viii. Raking bond

English bond is one of the strongest and most commonly used brick bonds. In this bond, alternate courses consist entirely of headers and stretchers.



- i. Header course: Bricks are laid with their short face (header) visible on the wall face.
- ii. Stretcher course: Bricks are laid with their long face (stretcher) visible.
- iii. A queen closer is placed next to the quoin header in the header course.
- iv. Vertical joints in successive courses are properly staggered to provide good interlocking.
- v. It is particularly suitable for load-bearing walls.

Important features of English bond:

- i. Provides high strength and stability.
- ii. Gives good load distribution.
- iii. Provides proper bonding/interlocking between courses.
- iv. Queen closers are used near the quoins in header courses.
- v. Avoids continuous vertical joints.
- vi. Commonly used in load-bearing masonry walls and foundations.

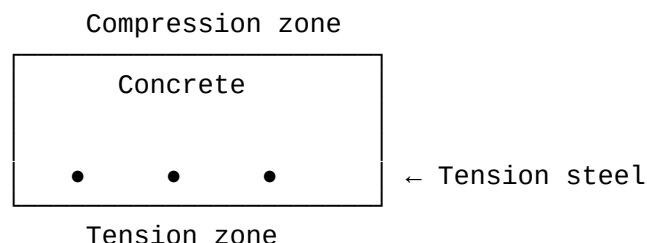
4. Describe singly reinforced section and doubly reinforced section of concrete structure particular for RCC beam section. State the main use of both sections and exemplify of both sections with sketches emphasizing on earthquake resistant need.

Answer:

An RCC beam is generally designed to resist bending moment and shear. Based on the provision of reinforcement, RCC beams are classified mainly as singly reinforced and doubly reinforced beams. Singly reinforced and doubly reinforced are the two methods of configuration of steel bars in RCC sections.

Singly Reinforced Beam:

Singly reinforced concrete section can be defined as the concrete section in which the main reinforcement bars are provided only in the tension zone (bottom part of beam section) as the concrete is weak in tension and prone to cracks and failures.



Singly reinforced concrete sections are used in construction works:

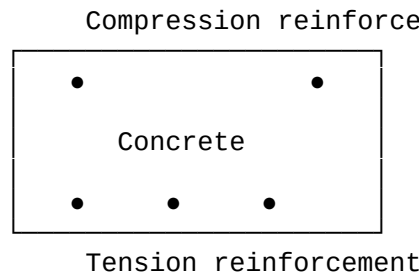
- when the Bending Moment is relatively small.
- Suitable when the depth of beam is not restricted.
- Commonly used in floor beams, slabs, beams, lintels and other ordinary structural members.

Limitation:

The beam depth cannot be reduced indefinitely because the compression capacity is mainly provided by concrete.

Doubly Reinforced Beam

Doubly reinforced concrete sections can be defined as those concrete sections in which the longitudinal steel bars are provided in both tension and compression zone (both top and bottom of the beam sections). Doubly reinforced section is provided when the heavy loads are to be subjected on them without increasing the depth of beam section due to structural and architectural constraint.



Doubly reinforced concrete sections are used in construction works:

- when the depth of beam is restricted and additional moment capacity is required.
- when the beam is subjected to reversal of stress such as earthquakes.
- to increase stiffness and reduce long term deflection.
- in heavily loaded structures such as bridges, multistorey buildings and retaining structures.

For earthquake-resistant RCC frames, beams should be detailed so that they undergo ductile flexural yielding rather than brittle failure.

- Reinforcement should be provided at both top and bottom faces of beams, especially in seismic moment-resisting frames.
- At beam-column joints, top reinforcement is important near supports because earthquake loading can reverse the bending moment.
- Adequate stirrups should be closely spaced near beam ends to prevent shear failure and provide confinement.
- Proper anchorage and development length must be provided.
- The design should follow the strong-column–weak-beam philosophy, so that plastic hinges form preferably in beams rather than columns.

5. **Define Hooke's law. Calculate the shear force and bending moment of the simply supported beam with span 4 m (c/c) and a point of 10 kN at the center of the span. Also draw the shear force and bending moment diagram.**

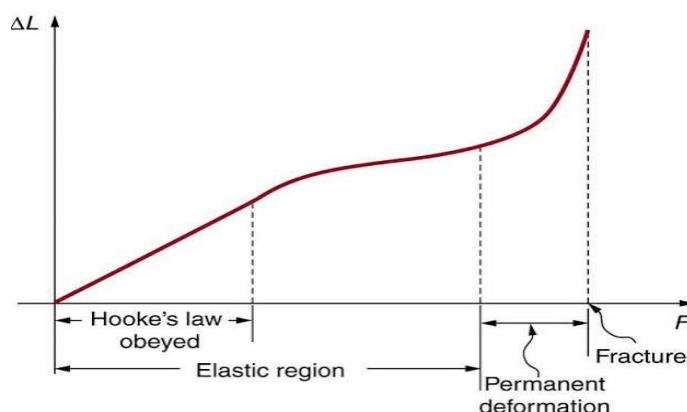
Answer:

Hooke's law states that, within the elastic limit, stress is directly proportional to strain.

$$\sigma \propto \epsilon$$

$$\text{or, } \sigma = E\epsilon$$

where E = Modulus of elasticity.



For a simply supported beam of span $L=4\text{m}$, carrying a point load $P=10\text{kN}$ at the center:

Given,

Span, $L=4\text{m}$

Central point load, $P= 10\text{kN}$

Load is at $L/2=2\text{m}$

i. Support reactions

By symmetry,

$$R_A=R_B= 2P$$

$$R_A=R_B= 5\text{kN}$$

ii. Shear Force

Taking upward shear as positive:

- From A to the load: $V= +5\text{kN}$
- Immediately after the 10 kN load: $V= 5-10= -5\text{kN}$
- From the load to B: $V= -5\text{kN}$

Therefore, the maximum shear force is:

$$V_{\max}= 5\text{kN}$$

iii. **Bending Moment**

The maximum bending moment occurs at the center:

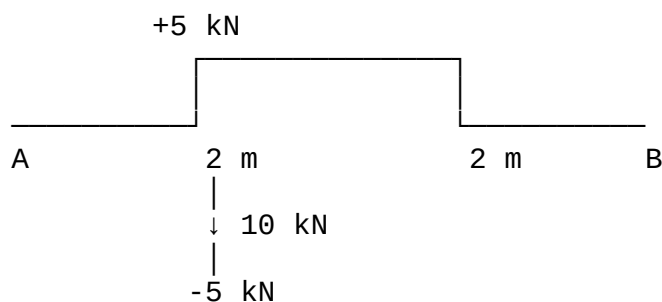
$$M_{\max}= R_A(2L)$$

$$M_{\max}=5(2)=10\text{kN}\cdot\text{m}$$

At both supports:

$$M_A=M_B=0$$

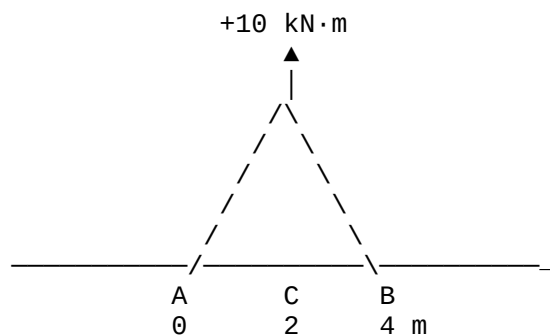
Shear Force Diagram (SFD)



Bending Moment Diagram (BMD)

The BMD is triangular because there is a central point load.

BMD



Location	Shear Force	Bending Moment
A (0 m)	+5 kN	0
Center (2 m)	+5 kN before load / -5 kN after load	+10 kN·m
B (4 m)	0 after reaction	0

$$R_A=R_B=5\text{kN } V_{\max}=5\text{kN } M_{\max}=10\text{kN}\cdot\text{m}$$

6. **What are the different kinds of urban housing? Describe in detail about “site and services”.**

Answer:

Urban housing refers to residential accommodation provided in urban areas to meet the housing needs of different income groups and household types.



Types of Urban Housing:

The major types are:

- i. **Individual/Detached Housing:** A separate house occupied by a single family, usually with its own plot and open space.
- ii. **Semi-detached Housing:** Two houses are joined by a common wall but have separate entrances and services.
- iii. **Row/Terraced Housing:** A series of houses are constructed side by side with common walls.
- iv. **Apartments/Flats:** Several dwelling units are accommodated in a multi-storey building with common facilities.
- v. **Group Housing:** A planned development containing several housing units with shared roads, open spaces and community facilities.
- vi. **Site-and-Services Housing:** Plots are provided with basic infrastructure and services, while occupants construct or improve their houses progressively.
- vii. **Rental/Social Housing:** Housing provided mainly for low- and middle-income households at affordable rents.
- viii. **Slum/Rehabilitation Housing:** Housing developed to improve or replace inadequate housing in slum and informal settlements.

Site-and-services housing is a housing development approach in which the government or development agency provides serviced residential plots along with basic infrastructure, while the beneficiaries construct their houses themselves, either initially or progressively according to their financial capacity.

It is particularly suitable for low- and middle-income groups because it reduces the initial cost of housing.

Main components:

i. Serviced Land:

Suitable land is subdivided into residential plots of appropriate size. The plots are generally provided at an affordable price or through subsidized schemes.

ii. Basic Infrastructure:

The development authority provides essential services such as:

- Access roads and footpaths
- Water supply
- Drainage and/or sewerage
- Electricity
- Street lighting
- Solid-waste collection facilities
- Other essential utility services

iii. Housing Construction:

The beneficiary constructs the house according to their financial capacity and family requirements. Construction may be carried out in stages.

iv. Community Facilities:

A good site-and-services scheme also reserves land for:

- Schools
- Health posts/clinics
- Markets
- Parks and playgrounds
- Community buildings
- Religious and cultural facilities

v. Financial Assistance:

Loans or other financial support may be provided to beneficiaries for house construction. This allows low-income families to gradually improve their houses.

Advantages of site and services housing:

- i. Affordable compared with fully constructed housing.
- ii. Provides planned residential areas.
- iii. Makes housing accessible to low-income groups.
- iv. Encourages self-help and community participation.
- v. Allows incremental construction according to income.

Disadvantages:

- i. Beneficiaries may not have sufficient funds to construct houses.
- ii. Houses may remain incomplete for long periods.
- iii. Poor construction quality may occur without technical supervision.
- iv. Land acquisition and infrastructure development require substantial initial investment.
- v. Provision of community facilities may be delayed.

7. What do you understand by the term “land use planning”? Write the importance of land planning in urban development which agencies are involving for urban development Nepal? [3+4]

Answer:

Land use planning is the systematic process of assessing, allocating and regulating land for different uses according to its physical characteristics, environmental conditions, socio-economic needs and development objectives. In urban development, land is planned and zoned for uses such as residential, commercial, industrial, institutional, recreational, agricultural, transportation and open spaces.



The importance of land use planning in urban development are listed below:

- i. **Optimum utilization of land**
Ensures that scarce urban land is used efficiently and appropriately.
- ii. **Planned urban growth**
Controls haphazard development and urban sprawl.
- iii. **Proper zoning**
Separates incompatible activities, such as heavy industries from residential areas.
- iv. **Provision of infrastructure**
Helps plan roads, water supply, drainage, sewerage, electricity and other services according to future land requirements.
- v. **Environmental protection**
Protects forests, agricultural land, wetlands, rivers, open spaces and other environmentally sensitive areas.
- vi. **Disaster risk reduction**
Prevents settlement in areas vulnerable to floods, landslides, earthquakes and other hazards.
- vii. **Improved transportation**
Coordinates residential, commercial and employment areas with road and public transportation networks.
- viii. **Provision of public facilities**
Ensures adequate land for schools, hospitals, parks, playgrounds, government buildings and other community facilities.
- ix. **Balanced development**
Promotes balanced development between different parts of the city and between urban and rural areas.
- x. **Improved quality of life**
Provides a healthy, safe and organized urban environment with adequate housing, services and open spaces.

8. **What is a periodic plan of a municipality? How can municipality play its roles in development of Nepal?**

Answer:

A periodic plan of a municipality is a medium-term development plan prepared by a municipality for a fixed period, generally five years, to guide the municipality's overall development. It identifies the municipality's

existing situation, problems, priorities, objectives, strategies, programs, projects, required resources and implementation mechanisms for the planned period. The periodic plan provides a link between the municipality's long-term vision and its annual programs and budgets.

Under Nepal's federal system, municipalities are important local governments and are responsible for delivering many basic services and promoting local development. Municipality plays important roles in development of Nepal by:

i. Local Infrastructure Development

Municipalities develop and maintain:

- Local roads and drainage
- Bridges and culverts
- Street lighting
- Public buildings
- Parks and public spaces

ii. Water Supply and Sanitation

They support and regulate:

- Drinking-water services
- Drainage and sanitation
- Solid-waste management
- Public toilets
- Local environmental sanitation

iii. Urban Planning and Land Use

Municipalities prepare and implement:

- Land-use plans
- Settlement development plans
- Building by-laws
- Building permit systems
- Urban development and zoning regulations

iv. Education and Health

They support basic-level education and local health services such as:

- Basic schools
- Health posts and basic health facilities
- Public-health and sanitation programs
- Nutrition and awareness programs

v. Local Economic Development

Municipalities can promote:

- Local industries and enterprises
- Agriculture and agro-based businesses
- Tourism
- Markets and commercial activities
- Employment and skill-development programs

9. **Write on vernacular architecture of Nepal in general and Kathmandu in particular.**

Answer:

Vernacular architecture refers to buildings designed and constructed using locally available materials, traditional construction techniques, local craftsmanship and cultural practices, generally without formal architectural design. It develops in response to the local climate, geography, available resources, social customs and lifestyle. Nepal has diverse vernacular architecture because of its varied Terai, Hill and Mountain environments.

i. Terai Region

Traditional houses in the Terai are generally constructed using: Mud, Bamboo, Timber, Thatch and locally available bricks

Features:

- Thick earthen walls provide thermal comfort.
- Sloping roofs are commonly used in areas with heavy rainfall.
- Courtyards and verandahs provide shade and ventilation.
- Locally available materials make construction relatively economical.



ii. Hill Region

Traditional hill houses commonly use: Stone masonry, Mud mortar, Timber, Slate or tile roofing

Features:

- Compact building forms are suitable for steep terrain.
- Thick masonry walls provide thermal insulation.
- Timber is used for floors, roofs and structural members.
- Houses are often adapted to the slope and local climate.



iii. Mountain Region

Traditional mountain buildings commonly use: Stone, Mud, Timber, Slate

Features:

- Very thick walls provide protection against cold temperatures.
- Small openings reduce heat loss.
- Compact forms reduce exposure to severe weather.
- Flat or gently sloping roofs are common in some high-altitude areas.



Kathmandu Valley has a distinctive architectural tradition developed by the Newar community, particularly in the historic settlements of Kathmandu, Patan and Bhaktapur.



Important characteristics

i. Traditional brick construction

- Locally produced fired bricks are extensively used.
- Brick masonry provides thermal mass and gives the buildings their characteristic appearance.

ii. Timber craftsmanship

- Timber is extensively used for beams, columns, doors, windows, balconies and roof structures.
- Intricately carved windows and doors are important features of Newari architecture.

iii. **Courtyard planning**

- Houses are commonly organized around an internal courtyard called chowk.
- Courtyards provide light, ventilation and communal space.

iv. **Multi-storey buildings**

- Traditional urban houses are generally narrow and vertically developed because of limited land availability.
- Ground floors were often used for commercial or storage purposes, while upper floors were used for living.

v. **Decorative façade**

- Exposed brick walls are combined with elaborately carved timber windows, doors and struts.
- The San Jhya (large traditional window) is a notable feature of Newari buildings.

vi. **Religious and cultural integration**

- Houses and settlements are closely associated with temples, monasteries, bahals, bahils, pati-pauwa, water spouts and other community spaces.
- Religious symbols and sculptures are incorporated into buildings.

10. **What do you understand by traditional architecture in Nepal? Illustrate three architecture landmarks in Nepal.**

Answer:

Traditional architecture in Nepal refers to the architectural forms, construction techniques, materials, decorative elements, and settlement patterns that have been developed and practiced by Nepali communities over generations. It reflects the country's culture, religion, climate, local materials, craftsmanship and social practices. Traditional architecture is characterized by the use of locally available materials such as brick, stone, timber, mud and clay tiles, together with traditional skills and construction methods.

Major Features:

- Use of local materials and indigenous construction techniques.
- Strong influence of Hinduism and Buddhism.
- Extensive use of carved timber, brickwork and stone sculpture.
- Courtyards, squares, narrow streets and community spaces are important elements.
- Buildings are adapted to local climate and topography.
- Rich ornamentation and symbolic architectural elements.

Three important architectural landmarks in Nepal:

- i. Nyatapoli Temple, Bhaktapur
- ii. Pashupati Temple, Kathmandu
- iii. Krishna Mandir, Patan

i. Nyatapoli Temple, Bhaktapur



- Located in Taumadhi Square, Bhaktapur.
- Built during the reign of King Bhupatindra Malla in the early 18th century.
- It is a famous example of Newar pagoda architecture.
- It has five storeys, making it one of Nepal's tallest traditional temples.
- The temple has a strong brick-and-timber structural system.
- It demonstrates the high level of traditional Newar craftsmanship and construction technology.

ii. Pashupatinath temple



- Located on the bank of the Bagmati River in Kathmandu.
- It is one of the most important Hindu temples in Nepal.
- The main temple is an excellent example of traditional Nepali pagoda architecture.
- It has two-tiered roofs, a gilded roof structure and richly decorated doors and architectural elements.
- Its architecture demonstrates the close relationship between religion, culture and traditional architecture in Nepal.

iii. Krishna Mandir, Patan



- Located in Patan Durbar Square, Lalitpur.
- Built during the reign of King Siddhi Narasimha Malla in the 17th century.
- Unlike the typical brick-and-timber pagoda, it is primarily constructed in stone.
- It is an important example of Shikhara-style architecture.
- Scenes from the Mahabharata and Ramayana are represented through its architectural and sculptural details.